MOSFET and CMOS Logic Circuits

- Definitions.
- MOS-based Inverter Design.
- The CMOS Inverter.
- CMOS Logic.
- Examples.

Digital Electronics

Digital electronics is the field of electronics that is concerned with the study of the performance and design of transistor-based circuits that perform logical functions. Usually, an elective course is given where basic and more advanced designs of circuits of logical gates are studied; first and second-order parameters are discussed. The study can also be extended to " digital system" circuit design.

In this course, only the very basics are discussed in the context of applications of MOSFETs as switching devices.

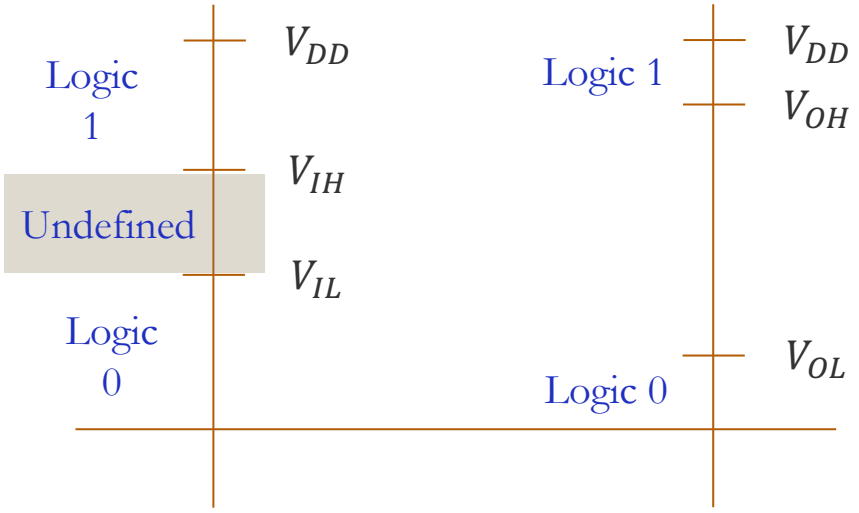
Before looking at specific designs, an introduction of basic definitions is given.

- Logic families
- Digital Signals.
- Noise Margins.
- Power dissipation.
- Switching/propagation times.
- Fan-in and fan-out.

Definitions

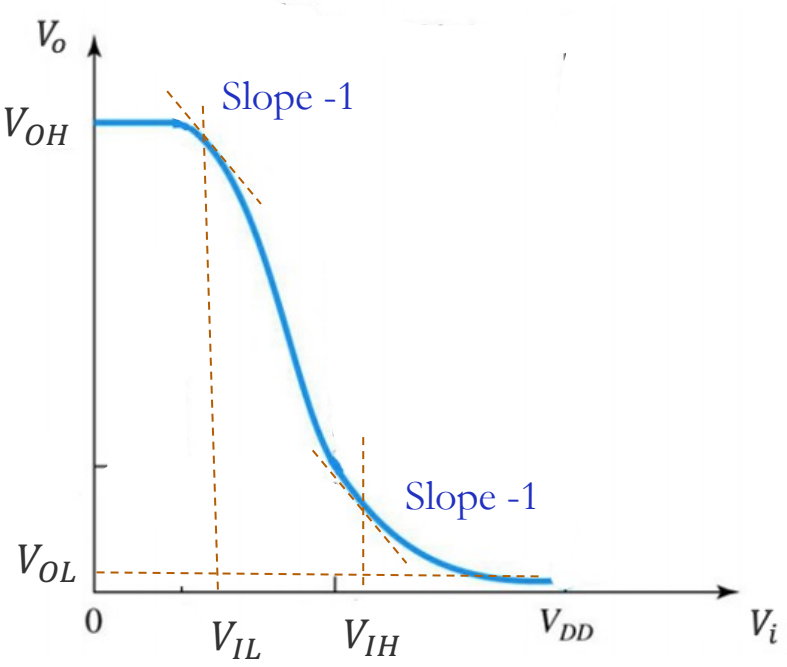
1) Logic families: Complimentary MOS (CMOS), Transistor-transistor Logic (TTL) and Hybrid Bipolar-CMOS (Bi-CMOS).

2) Digital Signals.



Input
Signal received by gate

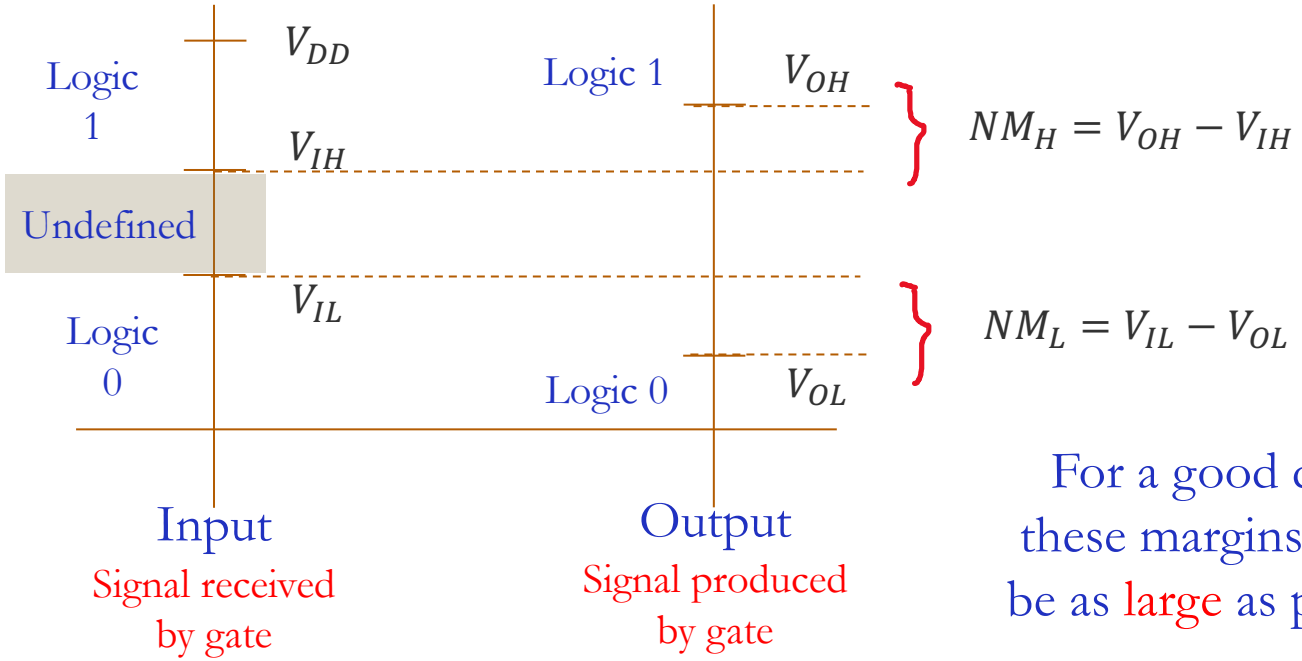
Output
Signal produced by gate



Basic inverter

Definitions

3) Noise Margins.



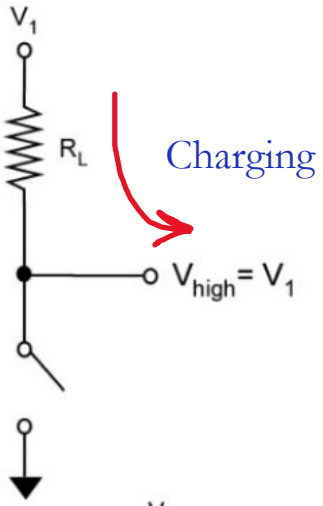
For a good design these margins should be as large as possible.

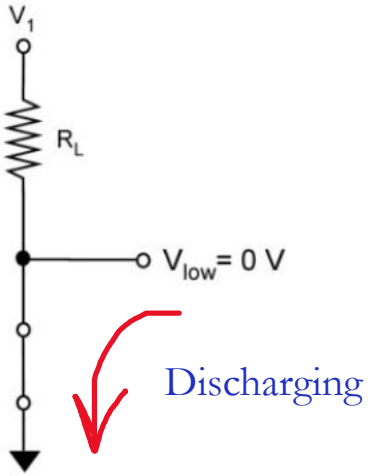
Example: In a certain technology, the gates produce a "1" with minimum voltage of 2.4 V and a "0" with maximum voltage of 0.4 V. Any input with maximum voltage of 0.8 is considered a "0" and minimum of 2 V for a "1". Find noise margins.

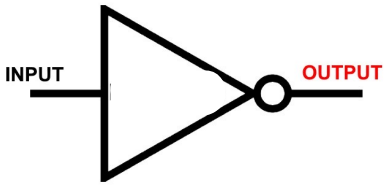
$$NM_L = V_{IL} - V_{OL} = 0.8 - 0.4 = 0.4\,V \qquad NM_H = V_{OH} - V_{IH} = 2.4 - 2 = 0.4\,V$$

Definitions

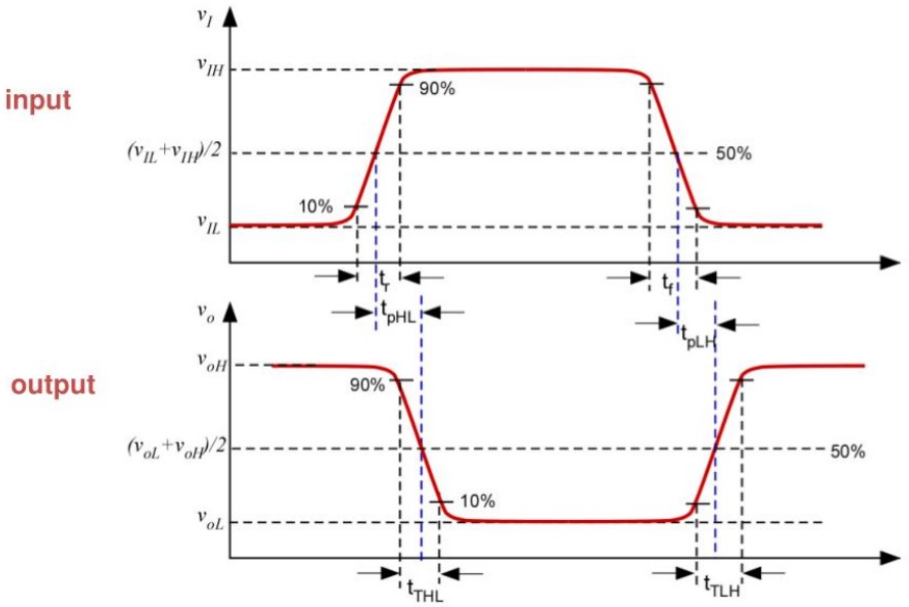
4) Switching/propagation times.



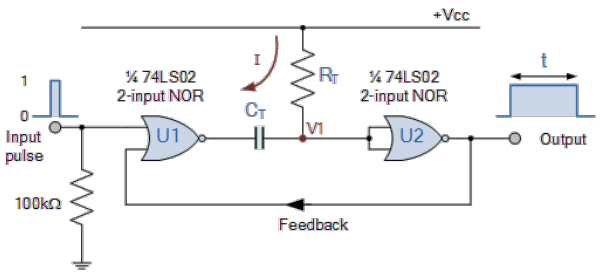




$$t_p = \frac{1}{2}(t_{pLH} + t_{pHL})$$



What is RC Delay?

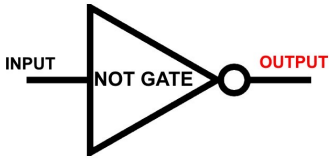


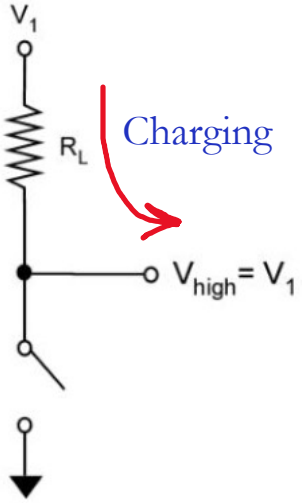
Definitions

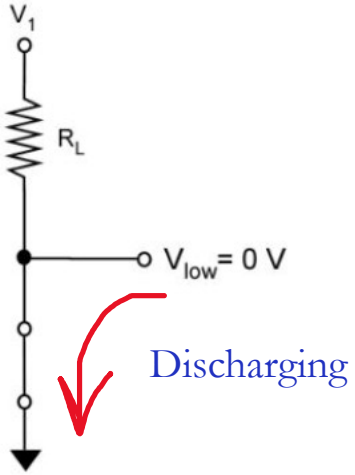
5) Power dissipation.

Static: the gate is idle.

Dynamic: during switching.





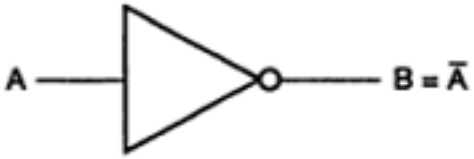


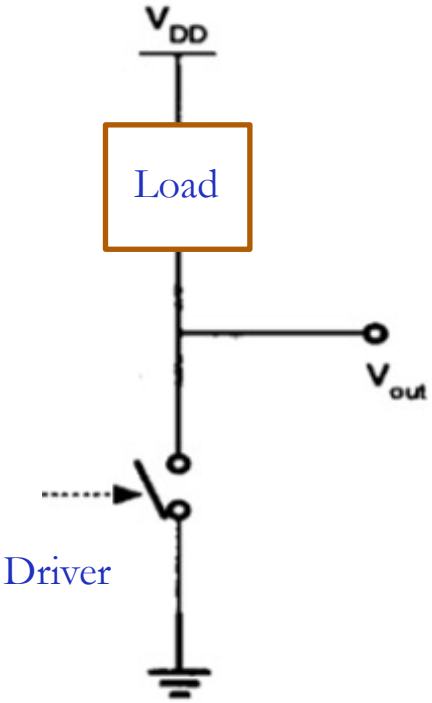
6) Fan-in and fan-out.

Fan-in: number of gates an output can drive (current).

Fan-out: number of inputs a gate can handle.

Basic Inverter Design

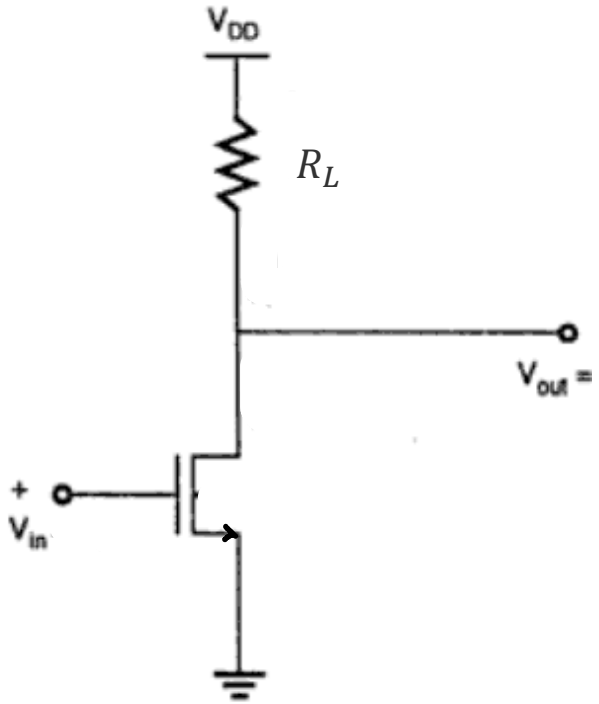


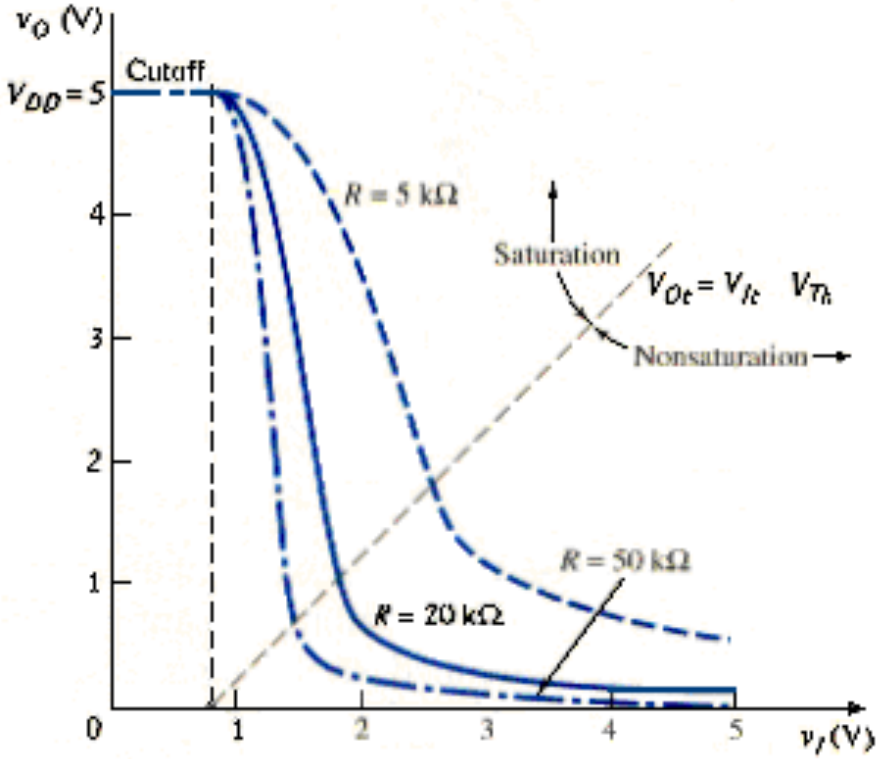


- Resistive load
- Enhancement NMOS load.
- Depletion NMOS load.
- CMOS technology.

Basic Inverter Design

1) Resistive load.

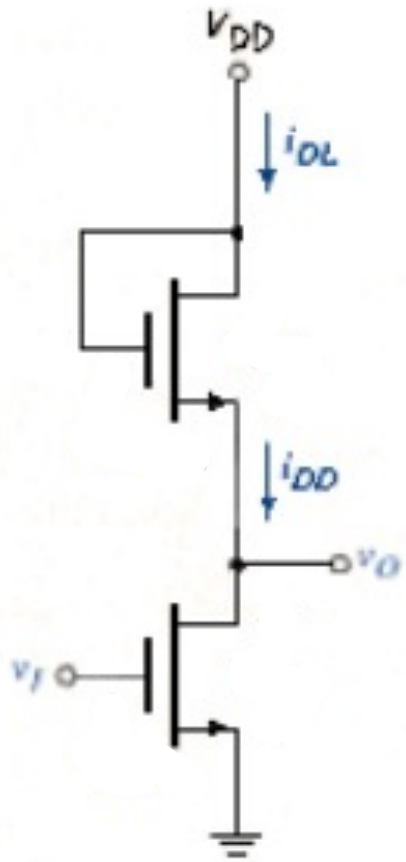


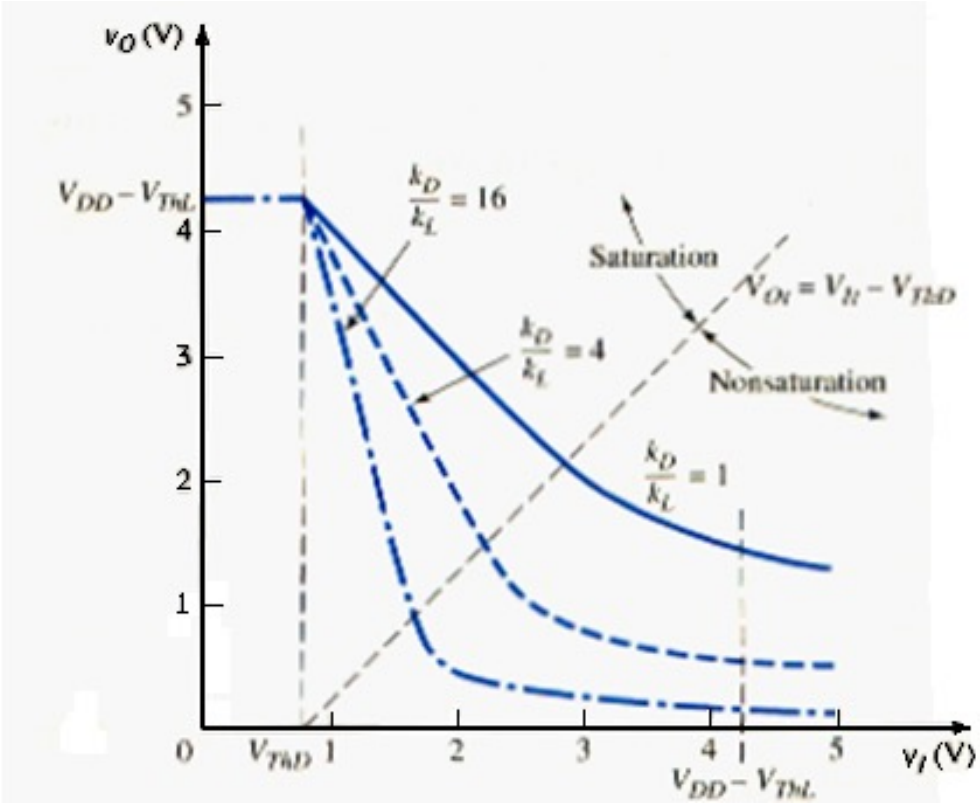


- Resistance is bulky and dissipative.
- V_{OL} depends on R_L (need high R).

Basic Inverter Design

2) Enhancement NMOS load.

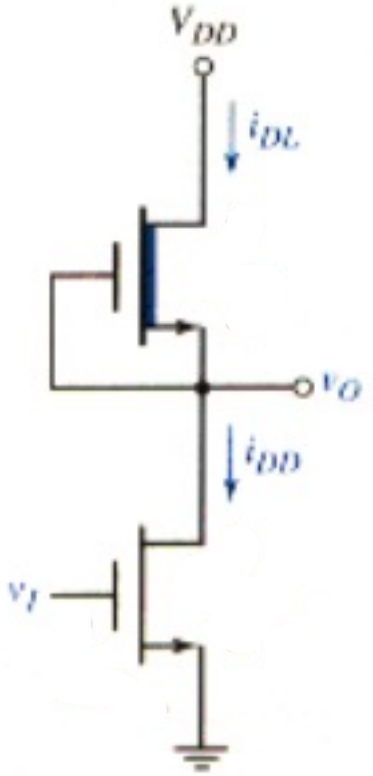


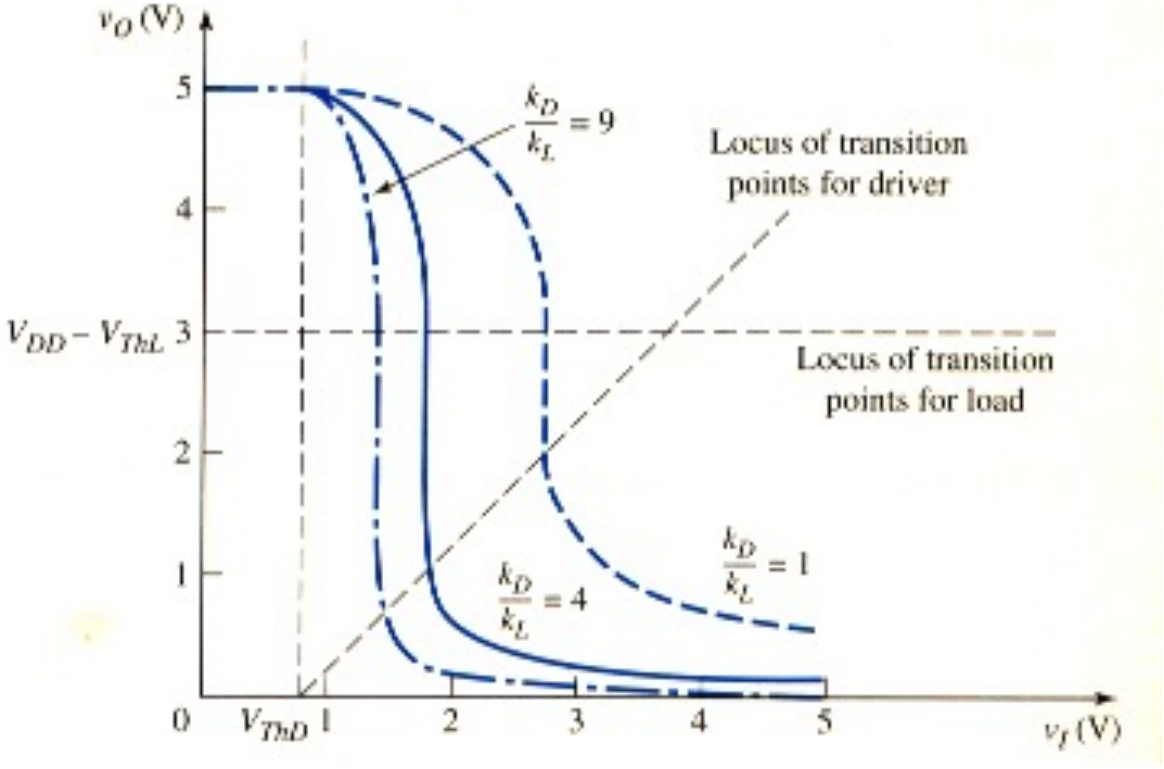


- V_{OH} is less than V_{DD}
- V_{OL} depends on size.

Basic Inverter Design

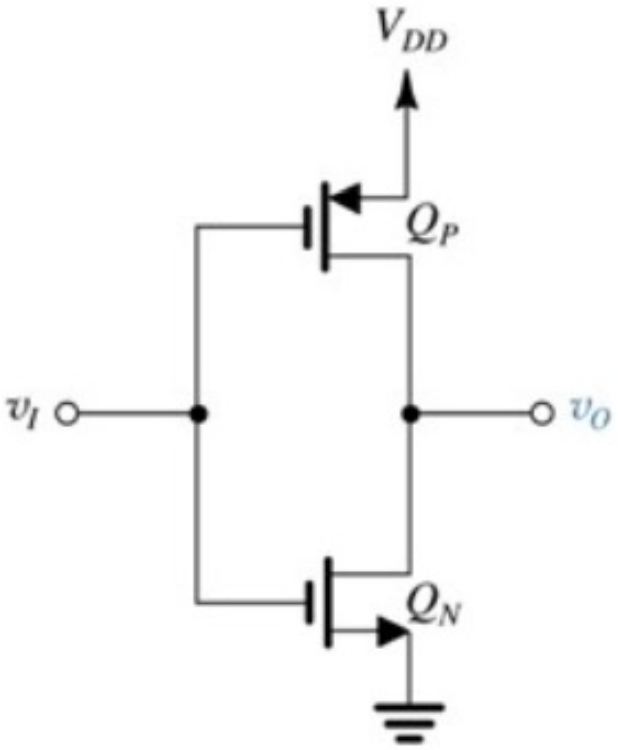
3) Depletion NMOS load.

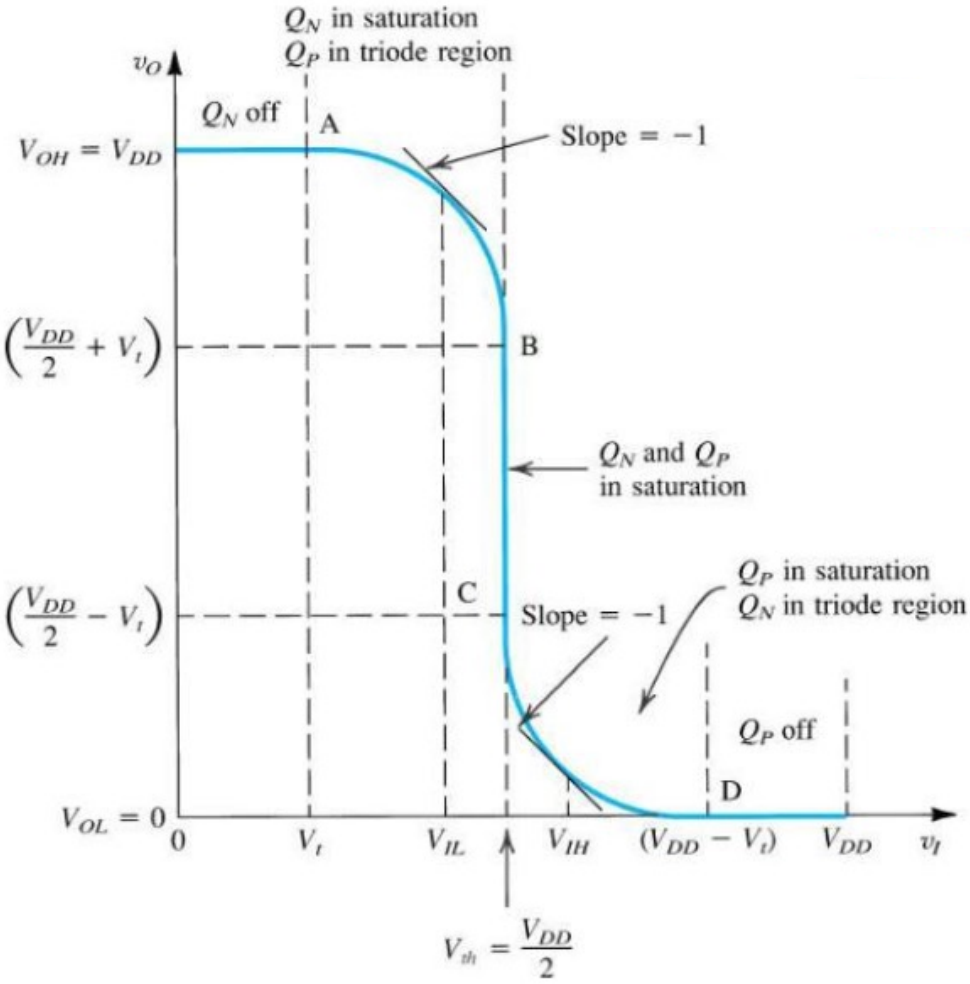




- V_{OL} depends on size.

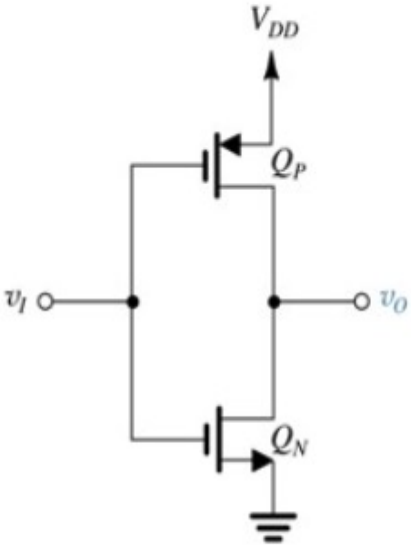
The CMOS Inverter

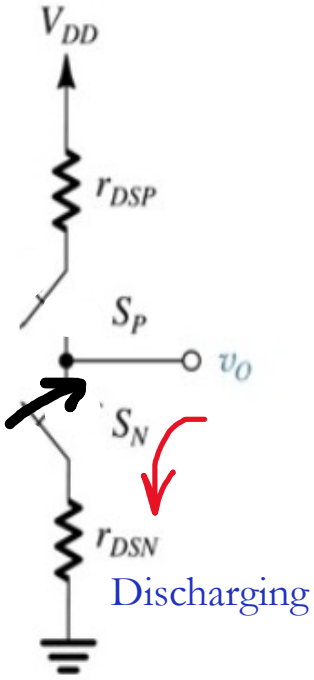




The CMOS Inverter

Input High





For the NMOS

$v_{gs} = v_i > V_{tN} \rightarrow$ ON

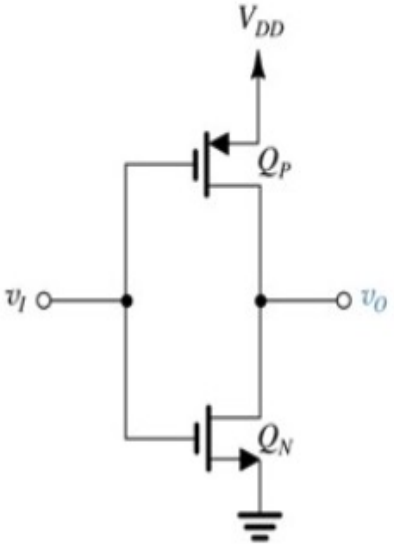
$v_o = 0$

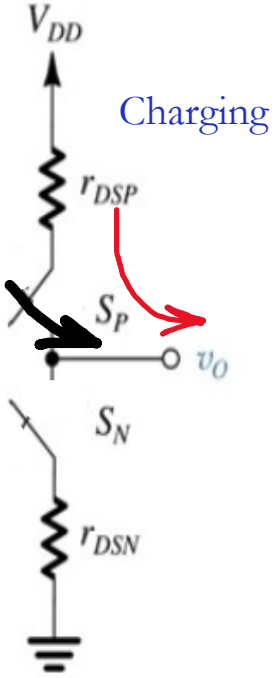
V_{OL}

For the PMOS

$v_{gs} = v_i - V_{DD} > V_{tP} \rightarrow$ OFF

Input Low





For the NMOS

$v_{gs} = v_i < V_{tN} \rightarrow$ OFF

$v_o = V_{DD}$

V_{OH}

For the PMOS

$v_{gs} = v_i - V_{DD} < V_{tP} \rightarrow$ ON

The Matched CMOS Inverter

In order to get the best noise margins for the CMOS inverter, the "driver" and the "load" need to be matched.

Calculation of V_{IL}

$$k_n(v_i - V_{tn})^2 =$$

$$k_p[2(v_i - V_{DD} - V_{tp})(v_o - V_{DD}) - (v_o - V_{DD})^2]$$

Take the derivative

$$2k_n(v_i - V_{tn}) =$$

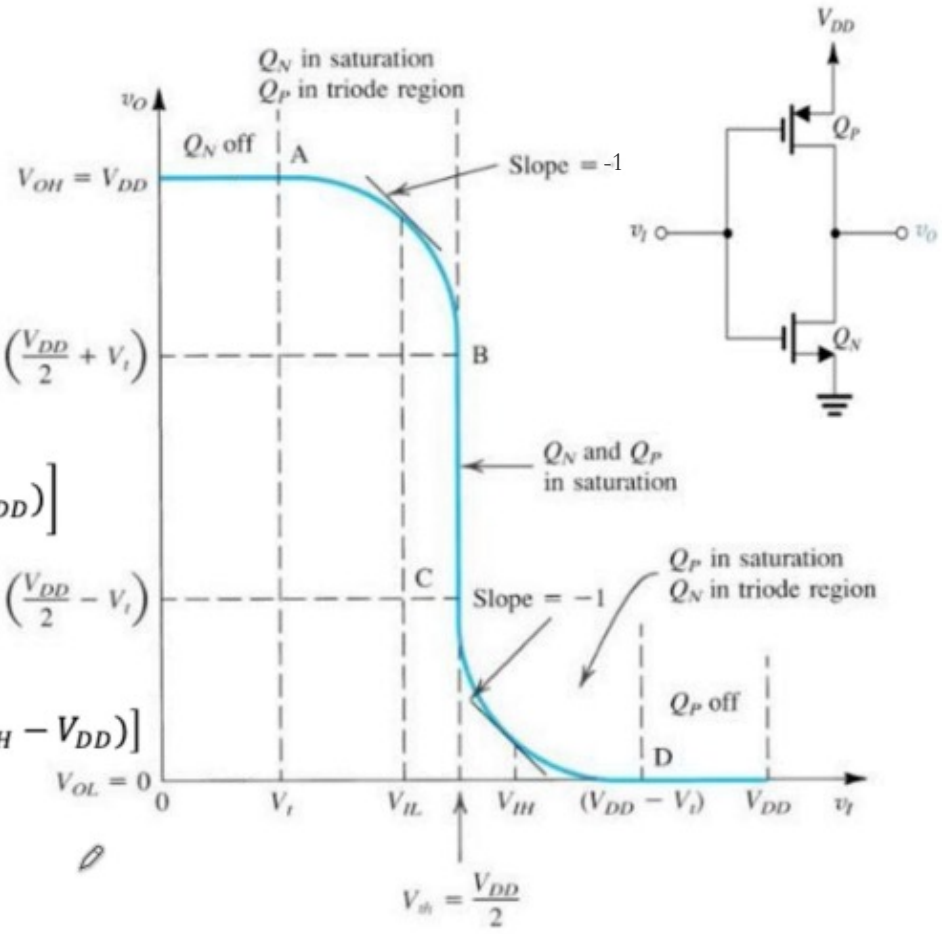
$$k_p\left[2(v_i - V_{DD} - V_{tp})\frac{dv_o}{dv_i} + 2(v_o - V_{DD}) - 2\frac{dv_o}{dv_i}(v_o - V_{DD})\right]$$

Derivative = −1

$$2k_n(V_{IL} - V_{tn}) =$$

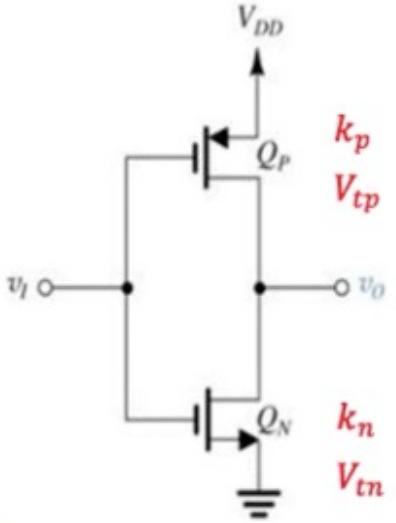
$$k_p[2(V_{IL} - V_{DD} - V_{tp})(-1) + 2(V_{OH} - V_{DD}) - 2(-1)(V_{OH} - V_{DD})]$$

Solve for V_{IL}. Repeat for V_{IH}.



The Matched CMOS Inverter

In order to get the best noise margins for the CMOS inverter, the "driver" and the "load" need to be matched.



Matching conditions

$$V_{tn} = |V_{tp}| = V_t$$

$$k_n = k_p \quad \rightarrow \quad \mu_n C_{ox}\left(\frac{W}{L}\right)_n = \mu_p C_{ox}\left(\frac{W}{L}\right)_p$$

$$\rightarrow \quad \left(\frac{W}{L}\right)_p = \frac{\mu_n}{\mu_p}\left(\frac{W}{L}\right)_n \rightarrow \quad \left(\frac{W}{L}\right)_p \approx 3\left(\frac{W}{L}\right)_n$$

So, matching is all about making the correct transistor sizes!

The gate length L is usually the same for all transistors on a chip (process factor)

Results

$$V_{IL} = \frac{1}{8}(3V_{DD} - 2V_t)$$

$$V_{IH} = \frac{1}{8}(5V_{DD} + 2V_t)$$

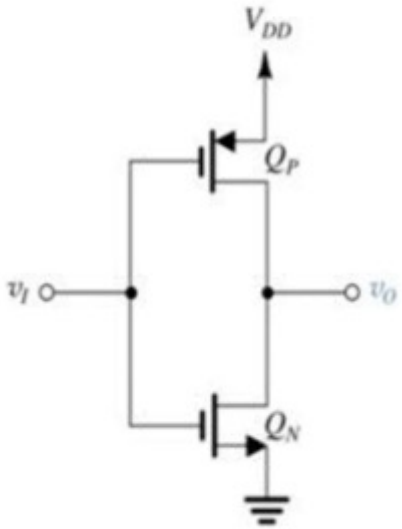
$$NM_H = NM_L = \frac{3}{8}V_{DD} + \frac{2}{3}V_t$$

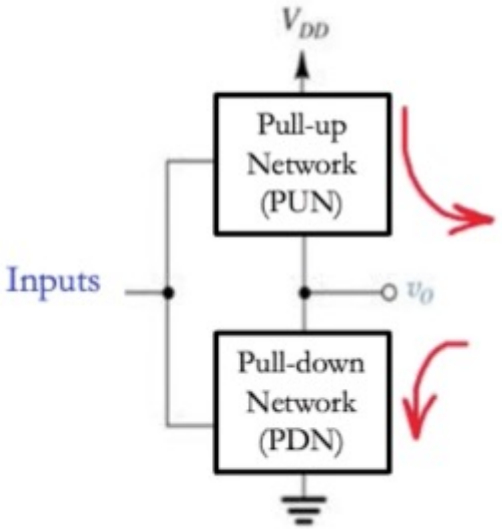
CMOS Logic Circuits

The CMOS digital circuits that we design should have (not only the correct logic outputs), but also the same properties as the basic inverter (or better):
Same logic levels, same noise margins.

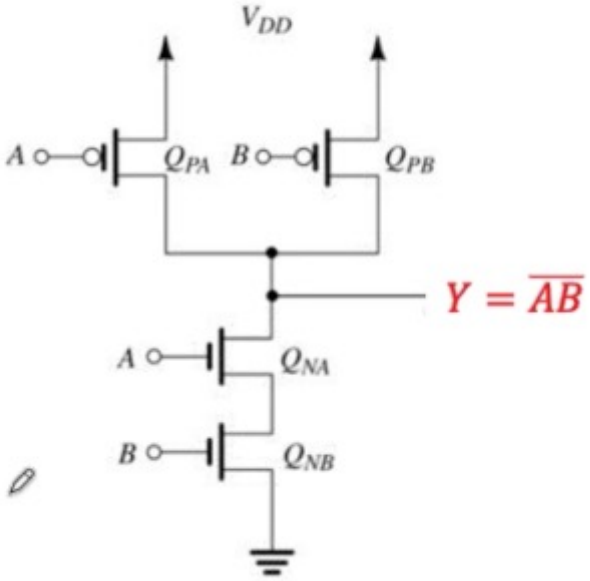
So, the properties of any CMOS logic circuit will always be referenced to the basic inverter.

First, how to get the correct logic function?



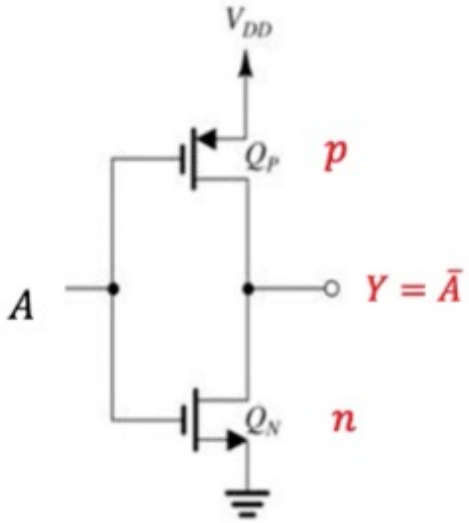


The NAND gate

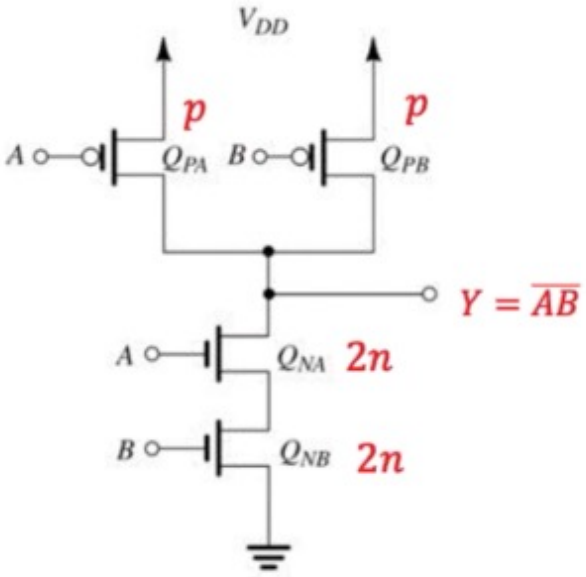


CMOS Logic Circuits

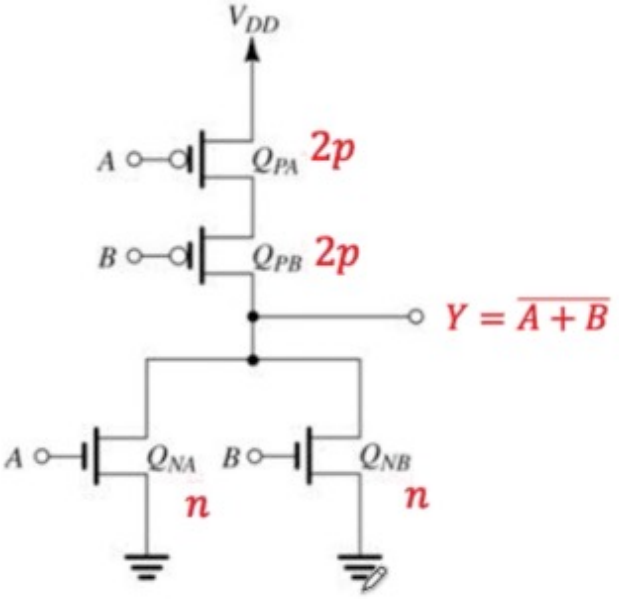
The NOT gate



The NAND gate



The NOR gate



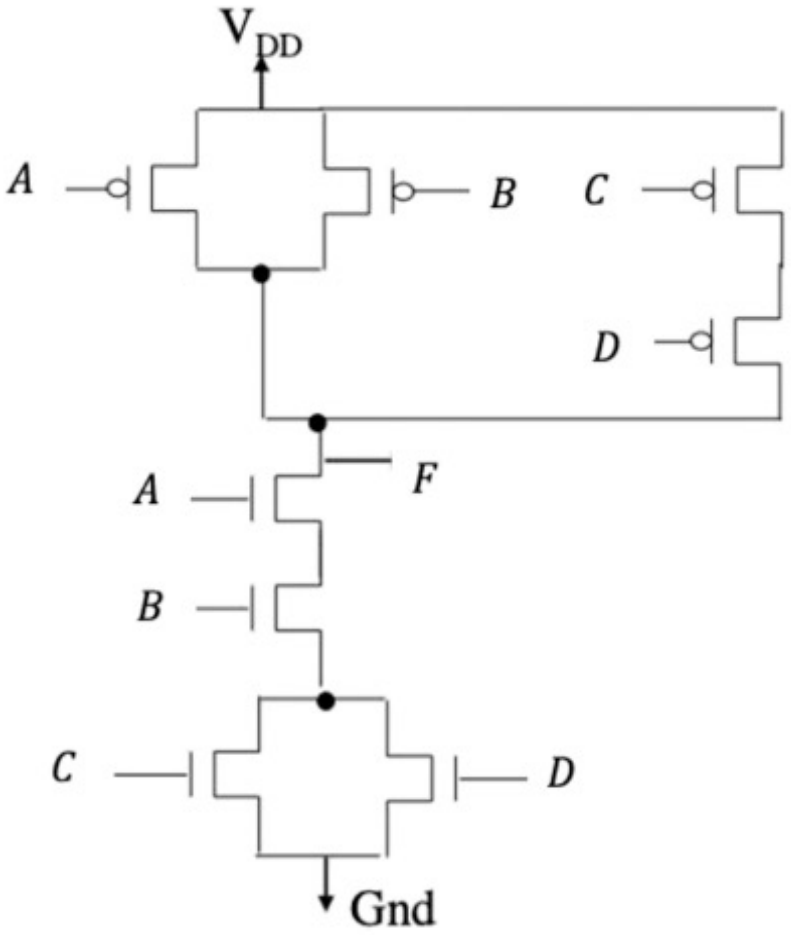
What about matching? Always refer to the basic inverter!

$$n = \left(\frac{W}{L}\right)_n \qquad p = \left(\frac{W}{L}\right)_p$$

$$\text{Circuit Area} = \sum(W \times L)_n + \sum(W \times L)_p$$
$$= \sum(p+n)\,L^2$$

Examples

1) What is the logic function implemented by the CMOS circuit below?



Examples

2) Implement the logical function $F = \bar{A}\,\bar{D} + \bar{B}\bar{C}\,\bar{D}$ using CMOS logic. Perform transistor sizing such that the properties match a CMOS inverter with $n = 1.5$, $p = 5$ and $L = 0.25\ \mu m$. Find the total circuit area.

$$F = \bar{\bar{F}} = \overline{\overline{\bar{A}\bar{D} + \bar{B}\bar{C}\bar{D}}}$$

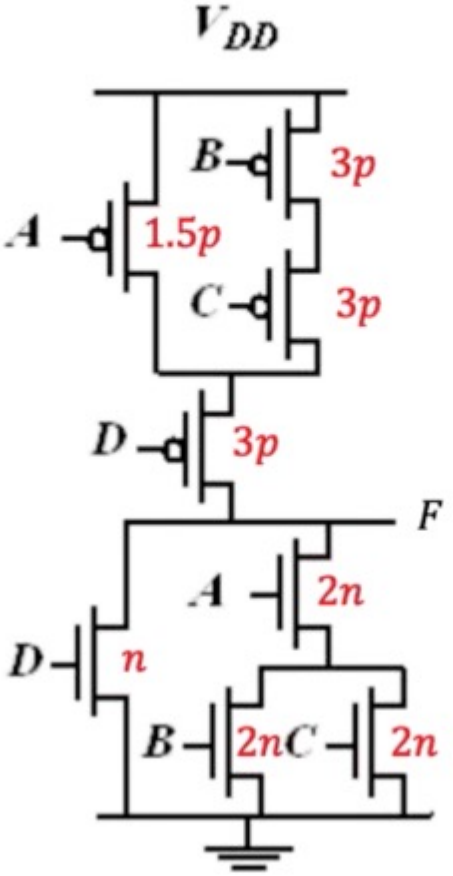
$$F = \overline{A(B + C) + D}$$

Circuit Area $= [p(1.5 + 3\times3) + n(1 + 3\times2)]L^2$

$$A = [(5)(10.5) + (1.5)(7)](0.25\ \mu m)^2$$

$$A = 63(0.25\ \mu m)^2$$

$$A = 3.94\ (\mu m)^2$$



Examples

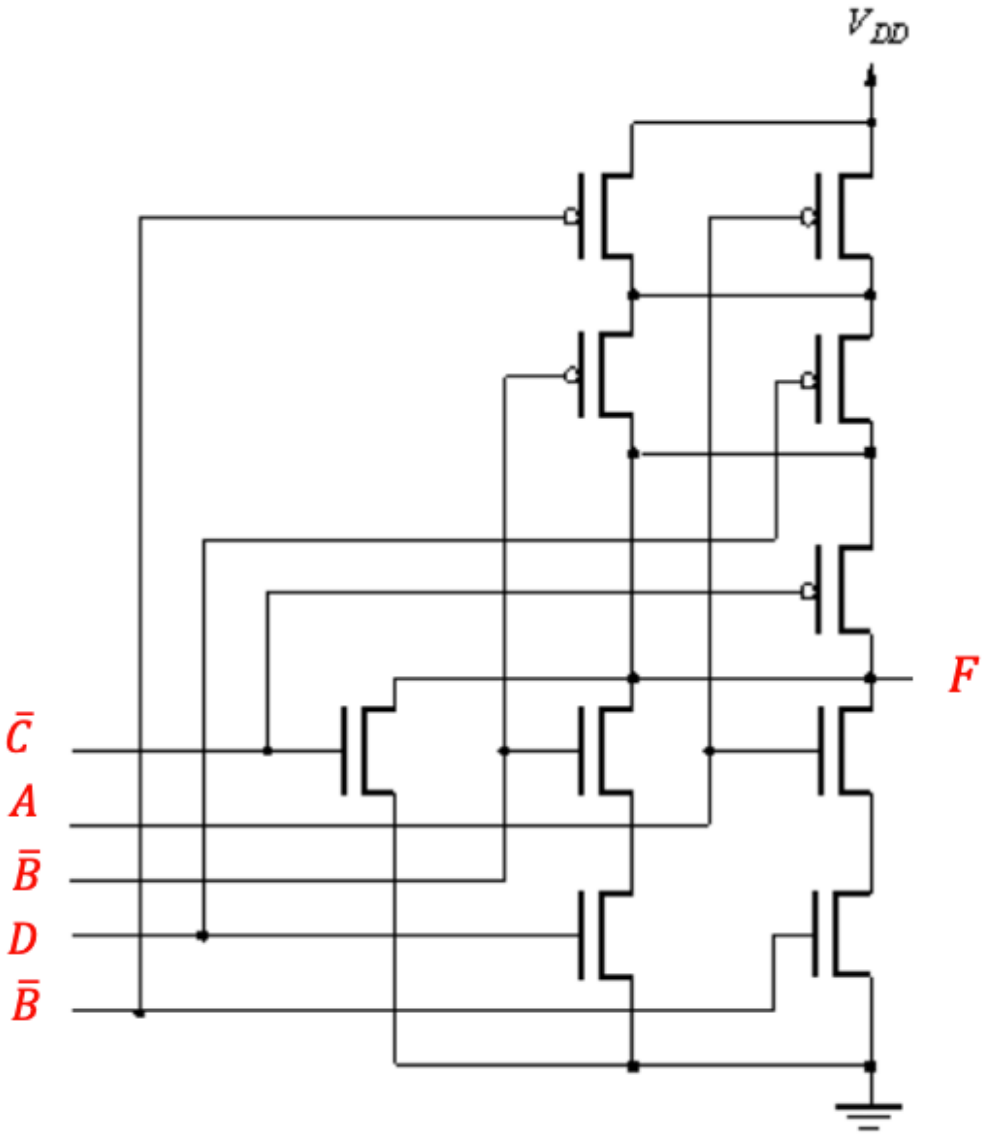
3) Implement the half-adder using CMOS logic. Perform transistor sizing such that the properties match a CMOS inverter with $n = 1.5$, $p = 5$ and $L = 0.25\ \mu m$. Find the total circuit area.

Examples

4) Find the logical function implemented by the given circuit.

Perform transistor sizing and find the total area such that the properties match a CMOS inverter with $p = 3$, $n = 1$ and $L = 0.2\ \mu m$.

For this particular logical function, does the given circuit represent the least area design? If not, produce a better CMOS implementation of the function. Find the savings in area.



Examples

5) Assuming that the input variables and their complements are available, design the digital function

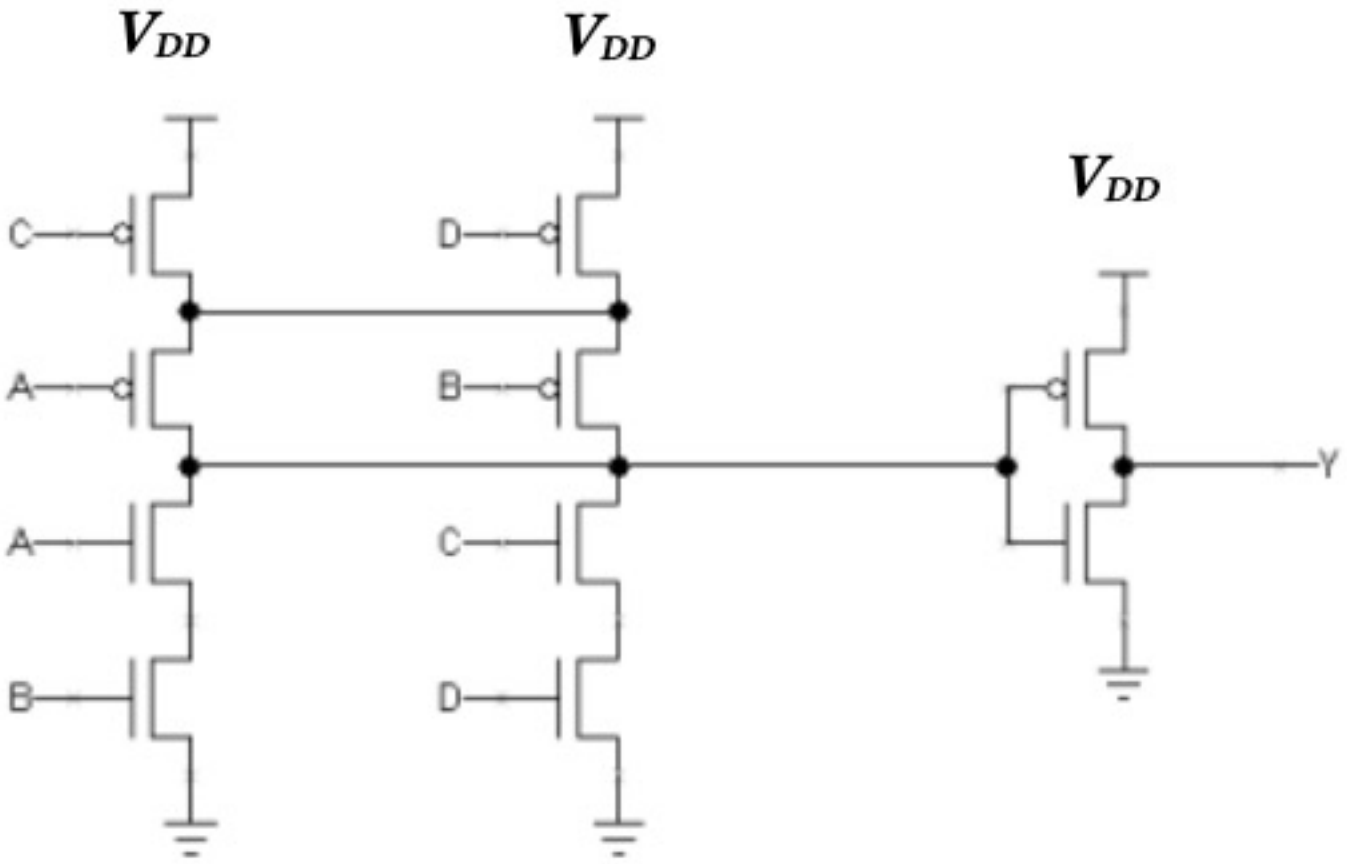
$$F(A,B,C) = \overline{A(B+\bar{C}) + \bar{B}(\bar{A}+\bar{C})}$$

using CMOS logic with a maximum of 6 NMOS transistors and 6 PMOS transistors.

Show the complete circuit, transistor sizing and total area. The reference inverter has $n = 1$, $p = 3$ and $L = 0.2\ \mu m$.

Examples

6) Redesign the circuit below such that the total chip area is less, but without compromising the current driving capability.



Q & A

Nano Photonics and Plasmonics Research

Send me an email if
you have any
questions.